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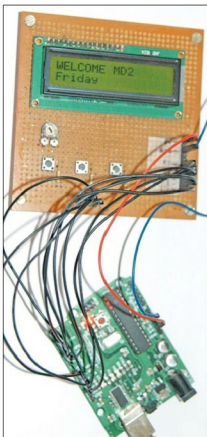


Fig. 2: Author's prototype

nected to 5V taken from Arduino. The switch-pressed states are read by digital read functions in the loop function of sketch.

The corresponding LCD display logic is also implemented in the loop function of sketch. Since switches S1 through S3 are tactile switches, contacts bounce during closure and release can cause false count. This problem is taken care of in the program by reading and comparing switch-closed conditions with a delay of 10ms.

Construction and testing

The circuit is wired on a general-purpose PCB. The prototype is shown in Fig. 2.

A USB A-B cable is used to upload the compiled sketch from the PC through Arduino IDE.

Photographs of some functions displayed on the LCD are shown in Fig. 3 and Fig. 4.

The operation is as follows:



Fig. 3: Multiplication result in mode 1



Fig. 4: Day of the week in mode 2

EFY Note

The source code (sketch) of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Press switch S3 repeatedly to change mode from 0, 1 and 2. Then, press S2 or S1 to change contents of the LCD.

The various functions of switches and messages displayed on the LCD are listed in Table I through Table III. For example, in mode 0, if you press switch S1 momentarily, Welcome will be displayed in the first line (row 1) of the LCD. At the same time, A a will be displayed in the second line (row 2) of the LCD. If you press S1 again, Welcome in the first row and B b will be displayed in the second row of the LCD. If you press switch S2, Welcome in the first row and A Apple will be displayed in the second row of the LCD.

VR1 preset can be used to adjust the contrast of the LCD. ●

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